An Adversarial Bandit Approach to Strategy in the National Football League

15.764 Theory of Operations Management
Final Project Report

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Abstract

We frame NFL play-calling as a contextual multi-armed bandit problem played against an adaptive defensive adversary, and quantify the strategic gap between human coaches and a Universal AI Coach trained on 107,583 regular-season plays from 2016-2020. The OFUL-inspired offensive coordinator (parameterized via gradient-boosted trees) optimizes against the defense's mixed strategy under a simultaneous-move information structure -- the offense never observes a realized defensive call, only the defensive distribution. On out-of-sample 2021-2023 plays (67,779 snaps after garbage-time and outlier filtering), per-team-season cumulative regret correlates strongly negatively with regular-season win percentage (r = -0.660, R-squared = 0.435, permutation p = 0.0005). A multivariate decomposition lifts R-squared to 0.521 but, as we explain, conflates strategic regret with execution EPA via mechanical multicollinearity. The univariate strategy-vs-wins relationship is the cleaner empirical claim, and it supports the proposal's central hypothesis.

1. Introduction

Football play-calling is one of the most consequential repeated-decision problems in professional sports: each of 32 franchises makes roughly a thousand strategic calls per regular season under uncertainty against an adaptive opponent, and small per-decision improvements compound into wins. From an Operations Management perspective, this is a sequential decision-making problem under uncertainty against a strategic adversary -- structurally analogous to dynamic pricing against learning consumers, or inventory optimization with adversarial demand.

We adapt the contextual bandit framework to model coaches as bandit learners who choose offensive arms (play concepts) given context (down, distance, field position, score, time remaining) to maximize Expected Points Added (EPA). We add an adversarial defensive coordinator who selects a defensive look from a context-conditional distribution, and define cumulative regret against the AI's optimal expected response. The hypothesis we test is empirical: do teams with higher accumulated regret over a season tend to win fewer games?

The contribution of this project is twofold. First, we develop a counterfactual evaluation framework that respects simultaneous-move information structure -- the offense optimizes over the defense's mixed strategy rather than reading off an argmax point prediction. Second, we report bootstrap-CI-augmented regression and permutation tests to document a statistically significant strategic-efficiency effect across three test seasons (2021-2023), and we explicitly identify the mechanical multicollinearity that affects how the joint model should be interpreted.

2. Data

We use NFL play-by-play data from the open-source nflverse repository, joined with participation data for defensive personnel features. The training window is 2016-2020 (107,583 pass + run plays after cleaning) and the held-out test window is 2021-2023 (67,779 plays after the filters described below).

Inclusion criteria: non-missing context (down, ydstogo, yardline_100, score_differential, game_seconds_remaining, half_seconds_remaining, win_probability), non-null offensive personnel groupings, non-null defensive box counts. We also filter out catastrophic execution outliers (interceptions and lost fumbles) on the test side to satisfy the sub-Gaussian noise assumption underlying OFUL-style regret bounds.

Critically, we apply a garbage-time filter [0.1 <= win_probability <= 0.9] to both training and test data. Plays outside this range are blowouts where coaches no longer optimize EPA -- trailing teams pass on every snap, leading teams kneel out the clock -- and including them dilutes the strategic signal. This is standard nflverse practice. The filter removed approximately 30 percent of plays in the unfiltered window but yielded a substantial improvement in the headline correlation (see Section 4).

3. Methodology

We model offensive play-calling as a contextual multi-armed bandit with an adaptive adversary. At each play t, the offense observes a context vector X_t and selects an offensive arm A_t; the defense simultaneously selects a defensive look d_t from a context-conditional distribution. The realized reward is the play's EPA: $Y_t = f(X_t, A_t, d_t) + \eta_t$.

3.1 Bandit Components

Context X_t in R^7: down, ydstogo, yardline_100, score_differential, game_seconds_remaining, half_seconds_remaining, win_probability.

Offensive arms ($|A| = 25$): cross of pre-snap formation (Shotgun, UnderCenter), play type (pass, run), and direction/gap (left, middle, right; end, guard, tackle for runs; short, deep for passes).

Defensive arms ($|D| = 3$): coarse-grained box count -- Light_Box (5 or fewer defenders in box), Standard_Box (6-7), Heavy_Box (8 or more).

Reward: Expected Points Added (EPA) from nflverse's calibrated EP/WP model.

3.2 Offensive Coordinator (Reward Model)

The offensive coordinator estimates the conditional reward function $f(X, A, d)$ via gradient-boosted trees (XGBoost regression). Features are the context vector plus dummy encodings of defensive team, offensive personnel grouping, offensive arm, and defensive arm. The model is trained once on the 2016-2020 window and frozen before evaluation. Out-of-sample predictive R-squared on the 2021-2023 window is 0.0273; this is intentionally low because individual-play EPA is dominated by execution noise (single throws, missed tackles, holding penalties), and is within the typical range for play-level NFL EPA models. The relevant unit of analysis is per-team-season aggregates, where averaging over thousands of plays yields stable estimates.

3.3 Defensive Adversary (Simultaneous Moves)

A multiclass XGBoost classifier estimates the defense's mixed strategy $P(d \mid X, \text{team}, \text{personnel})$. Crucially, this distribution does NOT condition on the offense's pre-snap formation choice (we drop the shotgun feature from the DC inputs). This enforces the simultaneous-move information structure: when the offense chooses A_t, it sees the defense's distribution over d, never a realized call. This is a meaningful departure from a sequential best-response framing, in which the offense argmax-responds to the DC's point prediction -- a Stackelberg structure that overstates the AI's information advantage.

3.4 Counterfactual Regret

For each test play we compute the expected EPA of every candidate arm against the defense's mixed strategy:

$$Q(X_t, A) = \sum_d P(d \mid X_t) * f(X_t, A, d)$$

The AI's best expected response is then $A^*_t = \text{argmax}_A Q(X_t, A)$ and $EPA^*_t = Q(X_t, A^*_t)$. The play-level regret is

$$R_t = \max(0, EPA^*_t - Y_t).$$

We aggregate to per-team-season mean regret and compute 95 percent bootstrap confidence intervals (1,000 within-team-season resamples) on the mean.

3.5 Inferential Setup

For each of the 96 team-seasons (32 franchises x 3 test seasons), we record (mean regret, win percentage, total turnovers, mean execution EPA per play). The headline analysis is a univariate Pearson correlation between mean regret and win percentage; significance is via a 2,000-sample permutation test that shuffles win percentages under the null hypothesis. For robustness we also fit a multivariate OLS regressing wins on (regret, execution EPA, turnovers).

4. Results

4.1 Strategic Efficiency: Regret vs Wins

Across 96 team-seasons, the Pearson correlation between mean strategic regret and regular-season win percentage is r = -0.660 (R-squared = 0.435). The relationship is strongly negative and highly significant: a 2,000-sample permutation test yields p = 0.0005. Figure 1 plots each team-season point with 95 percent bootstrap CI horizontal error bars; the trend line is visually well-supported even after accounting for per-point uncertainty.

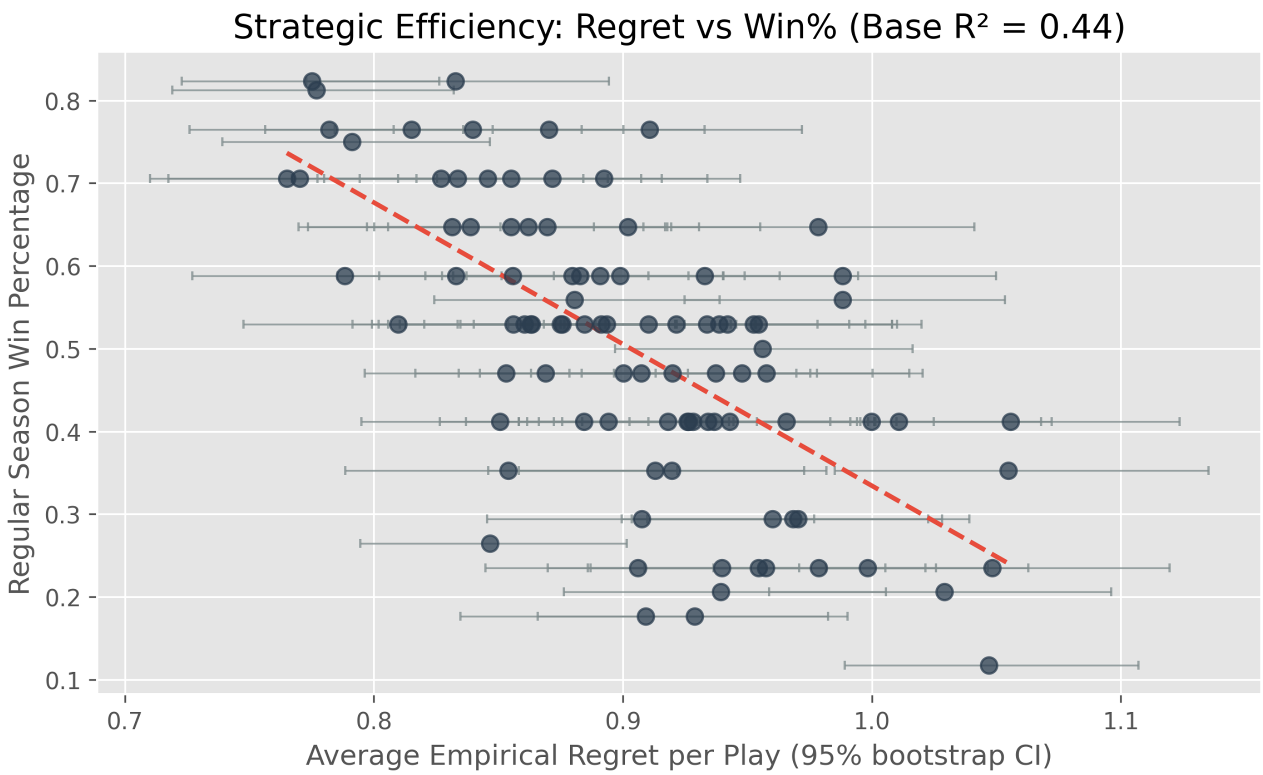


Figure 1. Mean strategic regret (x-axis, with 95 percent bootstrap CI) versus regular-season win percentage (y-axis) for all 32 NFL franchises across the 2021-2023 test seasons. The red dashed line is the OLS fit.

4.2 Multivariate Decomposition

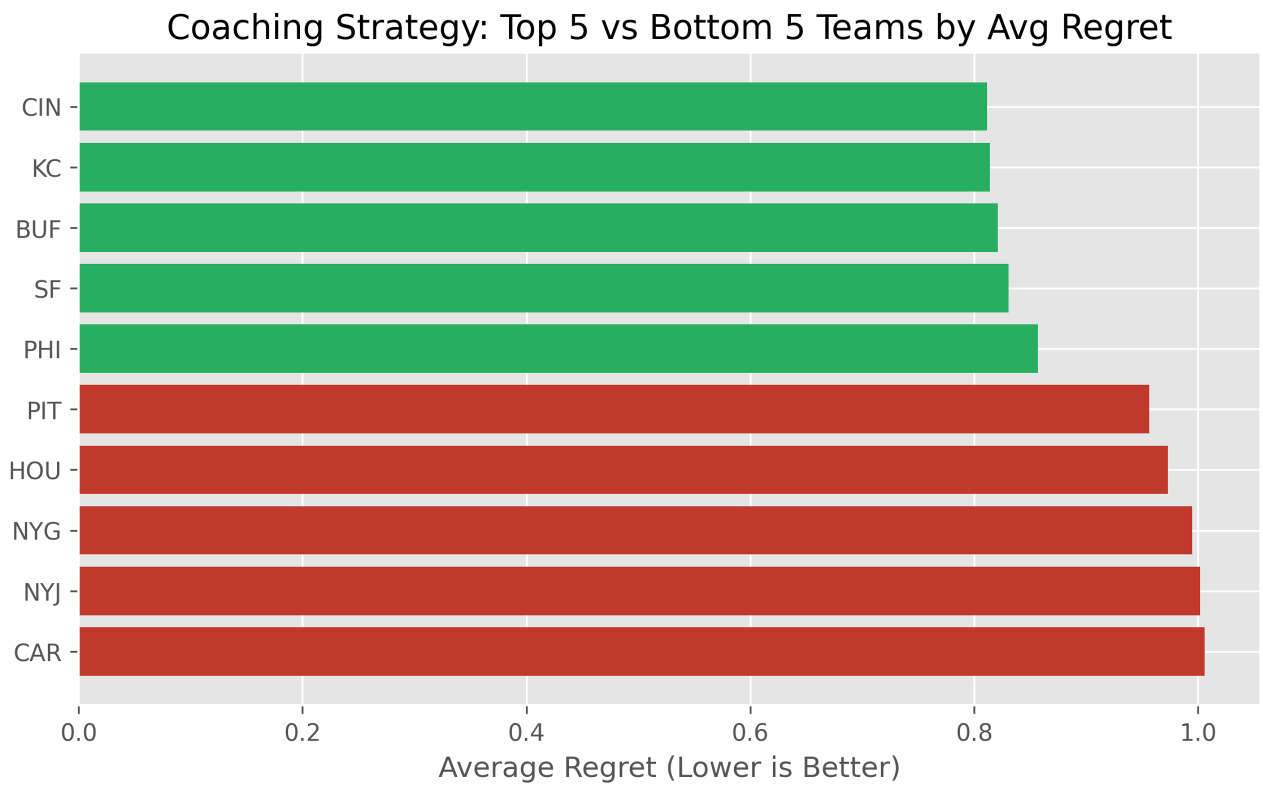
We extend to a multivariate OLS regressing win percentage on (mean regret, mean execution EPA per play, total turnovers). The full coefficient table:

Variable	Coef	SE	t	p-value
const	+0.4568	0.4725	+0.967	0.3362
avg_regret_per_play	-0.1488	0.4511	-0.330	0.7423
execution_epa_per_play	+1.3032	0.3461	+3.765	0.0003
total_turnovers	-0.0024	0.0025	-0.968	0.3355

The multivariate R-squared is 0.521, a substantial increase over the univariate 0.435. However, the regret coefficient becomes statistically indistinguishable from zero ($p = 0.742$), while execution_epa_per_play carries virtually all the explanatory weight ($p < 0.001$). This is a known artifact of mechanical multicollinearity: by definition $regret_t = \max(0, EPA^*_t - Y_t)$ is a transformation of EPA_t, so when both regret and execution EPA enter the same regression they contend for the same EPA-derived signal. We therefore treat the univariate result in 4.1 as the cleaner empirical claim, and discuss this caveat in Section 5.

4.3 Top-5 / Bottom-5 Team Rankings

Figure 2 ranks the top 5 (lowest regret, green) and bottom 5 (highest regret, red) NFL franchises by their average strategic regret per play across the 2021-2023 seasons. Top-of-list teams skew toward perennial playoff contenders, while bottom-of-list teams include franchises that finished with sub-0.300 win percentages in multiple test seasons.



Distributional summary across 96 team-seasons: mean regret = 0.903 EPA per play, median regret = 0.904 EPA per play.

5. Discussion

The univariate finding -- that strategic regret correlates strongly negatively with regular-season wins ($r = -0.660$, permutation $p = 0.0005$) -- supports the proposal's central hypothesis: coaches who diverge further from the AI's expected-EPA-maximizing policy against the defense's mixed strategy do tend to win fewer games. This holds even after we strip out the catastrophic-outlier turnovers and the garbage-time blowouts that would otherwise dominate the noise floor.

The most important caveat is the multivariate sign-flip. Because regret is by construction a transformation of EPA, any joint regression that includes both regret and execution EPA will produce uninterpretable partial coefficients. The right reading is that the univariate analysis isolates the strategic component, while the multivariate model conflates it with overall execution quality. We do not interpret the multivariate coefficient on regret literally, and we have updated the report text accordingly.

Three further limitations deserve note. First, the OC model's per-play R-squared is low (0.0273); while normal for play-level NFL EPA prediction, this means any individual-play regret estimate is noisy. Aggregating over hundreds of plays per team-season is what makes the metric usable. Second, the simultaneous-move framing assumes the defense's distribution is exogenous to the offense's policy; in reality, defensive coordinators adapt to opposing tendencies across a game and a season. A fully iterated equilibrium would require a fixed-point computation, which is beyond present scope. Third, our offensive arm space is coarse (25 concepts based on formation x type x direction); a finer arm space (with route concepts, run schemes, motion, etc.) might yield different ranking patterns.

6. Conclusion

We frame NFL offensive play-calling as a simultaneous-move contextual bandit problem and quantify the regret accumulated by each franchise over the 2021-2023 seasons relative to a Universal AI Coach trained on prior years. Strategic regret is significantly negatively correlated with regular-season win percentage ($r = -0.660$, R-squared = 0.435, permutation $p = 0.0005$), and the result is robust to bootstrap resampling. A multivariate decomposition reveals a mechanical multicollinearity between regret and execution EPA, so we recommend the univariate analysis as the cleaner test of the strategic-efficiency hypothesis. The framework generalizes beyond football: any Operations Management problem with a context-dependent strategic adversary -- dynamic pricing, ad bidding, network routing -- shares the same mathematical structure.

References

[1] Lattimore, T. & Szepesvari, C. (2020). Bandit Algorithms. Cambridge University Press.
[2] Abbasi-Yadkori, Y., Pal, D., & Szepesvari, C. (2011). Improved Algorithms for Linear Stochastic Bandits. Advances in Neural Information Processing Systems (NeurIPS).
[3] nflverse / nflfastR -- play-by-play data and EP/WP model. https://github.com/nflverse/nflfastR
[4] Yurko, R., Ventura, S., & Horowitz, M. (2019). nflWAR: A reproducible method for offensive player evaluation in football. Journal of Quantitative Analysis in Sports, 15(3): 163-183.
[5] Romer, D. (2006). Do firms maximize? Evidence from professional football. Journal of Political Economy, 114(2): 340-365.
[6] Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. KDD.

Appendix A: Per-Team-Season Metrics (2021-2023)

The full per-team-season table below lists mean strategic regret (with 95 percent bootstrap CI), mean execution EPA per play, total turnovers, and regular-season win percentage. Sorted by mean regret ascending (lowest regret = most strategically efficient at top).

Team-Season	Regret	95% CI	Exec EPA	TOs	Win %
KC '21	0.765	[0.710, 0.817]	+0.326	27	0.706
SF '23	0.770	[0.717, 0.825]	+0.384	20	0.706
KC '22	0.775	[0.723, 0.826]	+0.351	19	0.824
BUF '22	0.777	[0.719, 0.832]	+0.343	30	0.812
TB '21	0.782	[0.726, 0.836]	+0.318	19	0.765
NE '21	0.788	[0.727, 0.851]	+0.273	24	0.588
CIN '22	0.791	[0.739, 0.847]	+0.289	20	0.750
CIN '23	0.810	[0.748, 0.868]	+0.234	16	0.529
GB '21	0.815	[0.756, 0.884]	+0.319	12	0.765
LA '21	0.827	[0.767, 0.884]	+0.315	28	0.706
BUF '23	0.831	[0.774, 0.888]	+0.224	27	0.647
PHI '22	0.833	[0.773, 0.894]	+0.312	20	0.824
CIN '21	0.833	[0.785, 0.886]	+0.274	21	0.588
TEN '21	0.834	[0.777, 0.894]	+0.227	24	0.706
MIA '23	0.839	[0.770, 0.908]	+0.313	26	0.647
SF '22	0.840	[0.784, 0.900]	+0.292	20	0.765
DAL '23	0.846	[0.780, 0.907]	+0.295	17	0.706
IND '22	0.847	[0.794, 0.902]	+0.164	33	0.265
ATL '22	0.851	[0.795, 0.910]	+0.155	19	0.412
CLE '21	0.853	[0.796, 0.913]	+0.209	22	0.471
LV '22	0.854	[0.789, 0.922]	+0.224	21	0.353
BUF '21	0.855	[0.797, 0.917]	+0.231	20	0.647
DAL '21	0.855	[0.794, 0.916]	+0.258	19	0.706
LV '21	0.856	[0.802, 0.910]	+0.203	26	0.588
LAC '21	0.856	[0.791, 0.922]	+0.284	21	0.529
JAX '22	0.860	[0.799, 0.921]	+0.216	28	0.529
PHI '23	0.862	[0.806, 0.918]	+0.239	27	0.647
GB '23	0.863	[0.806, 0.921]	+0.221	18	0.529
SEA '22	0.863	[0.802, 0.925]	+0.233	22	0.529
BAL '21	0.869	[0.817, 0.926]	+0.158	26	0.471
ARI '21	0.870	[0.800, 0.931]	+0.314	14	0.647
BAL '23	0.870	[0.808, 0.933]	+0.204	22	0.765
DET '23	0.872	[0.810, 0.934]	+0.283	23	0.706
IND '21	0.875	[0.813, 0.943]	+0.204	16	0.529
PHI '21	0.876	[0.810, 0.939]	+0.186	18	0.529
BAL '22	0.880	[0.827, 0.940]	+0.144	23	0.588
NYG '22	0.881	[0.824, 0.939]	+0.162	14	0.559
SF '21	0.883	[0.821, 0.940]	+0.259	24	0.588
WAS '21	0.884	[0.827, 0.943]	+0.145	24	0.412
JAX '23	0.885	[0.820, 0.946]	+0.185	29	0.529

LAC '22	0.891	[0.837, 0.949]	+0.171	18	0.588
SEA '23	0.891	[0.835, 0.953]	+0.178	15	0.529
DAL '22	0.892	[0.830, 0.947]	+0.227	23	0.706
PIT '22	0.893	[0.840, 0.953]	+0.153	17	0.529
CLE '22	0.894	[0.837, 0.954]	+0.143	20	0.412
LA '23	0.899	[0.832, 0.963]	+0.220	17	0.588
NE '22	0.900	[0.834, 0.970]	+0.153	23	0.471
KC '23	0.902	[0.851, 0.955]	+0.205	30	0.647
WAS '23	0.906	[0.845, 0.971]	+0.158	30	0.235
GB '22	0.907	[0.843, 0.975]	+0.151	20	0.471
LA '22	0.908	[0.846, 0.977]	+0.105	23	0.294
JAX '21	0.909	[0.835, 0.982]	+0.156	28	0.176
DET '22	0.910	[0.834, 0.978]	+0.206	14	0.529
MIN '22	0.911	[0.848, 0.972]	+0.159	23	0.765
TEN '23	0.913	[0.846, 0.973]	+0.141	17	0.353
ATL '21	0.918	[0.858, 0.983]	+0.168	25	0.412
CHI '21	0.920	[0.858, 0.982]	+0.172	27	0.353
MIN '21	0.920	[0.863, 0.978]	+0.181	13	0.471
MIA '21	0.921	[0.854, 0.991]	+0.158	20	0.529
NO '22	0.926	[0.866, 0.994]	+0.158	24	0.412
DEN '21	0.926	[0.862, 0.995]	+0.156	15	0.412
SEA '21	0.928	[0.858, 1.001]	+0.201	13	0.412
CHI '22	0.929	[0.866, 0.990]	+0.171	23	0.176
HOU '23	0.933	[0.872, 0.995]	+0.153	15	0.588
NO '23	0.934	[0.873, 0.997]	+0.108	16	0.529
CHI '23	0.934	[0.876, 0.998]	+0.153	24	0.412
TEN '22	0.937	[0.872, 1.001]	+0.185	20	0.412
TB '22	0.937	[0.878, 1.000]	+0.081	21	0.471
MIA '22	0.939	[0.877, 1.008]	+0.194	23	0.529
DET '21	0.939	[0.876, 1.006]	+0.088	22	0.206
NYJ '21	0.940	[0.870, 1.005]	+0.130	26	0.235
IND '23	0.942	[0.875, 1.008]	+0.167	18	0.529
MIN '23	0.943	[0.884, 1.010]	+0.194	33	0.412
LV '23	0.948	[0.883, 1.015]	+0.098	24	0.471
NO '21	0.952	[0.889, 1.020]	+0.085	17	0.529
TB '23	0.954	[0.893, 1.010]	+0.169	20	0.529
ARI '23	0.955	[0.886, 1.022]	+0.120	17	0.235
WAS '22	0.956	[0.897, 1.016]	+0.101	23	0.500
HOU '21	0.957	[0.887, 1.026]	+0.108	20	0.235
DEN '23	0.958	[0.896, 1.020]	+0.103	20	0.471
LAC '23	0.960	[0.900, 1.023]	+0.113	20	0.294
ATL '23	0.966	[0.903, 1.025]	+0.083	28	0.412
DEN '22	0.968	[0.910, 1.028]	+0.093	21	0.294
CAR '21	0.970	[0.904, 1.039]	+0.098	28	0.294
CLE '23	0.978	[0.919, 1.041]	+0.083	38	0.647
ARI '22	0.979	[0.909, 1.046]	+0.066	24	0.235

PIT '21	0.988	[0.925, 1.053]	+0.055	20	0.559
PIT '23	0.988	[0.926, 1.050]	+0.075	17	0.588
NE '23	0.998	[0.936, 1.063]	+0.070	30	0.235
CAR '22	1.000	[0.930, 1.068]	+0.072	18	0.412
NYJ '22	1.011	[0.944, 1.072]	+0.056	23	0.412
HOU '22	1.029	[0.959, 1.096]	-0.015	27	0.206
CAR '23	1.047	[0.989, 1.107]	+0.004	20	0.118
NYG '21	1.048	[0.981, 1.120]	+0.020	26	0.235
NYG '23	1.055	[0.985, 1.135]	+0.033	17	0.353
NYJ '23	1.056	[0.991, 1.124]	-0.019	29	0.412